

A new fuzzy tri-objective model for a home health care problem with green ambulance routing and congestion under uncertainty

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ABSTRACT

With the increasing number of older people and high hospital treatment costs, older people tend to receive cures at home and more demand for home health care centers (HHCCs). Therefore, HHCCs should use appropriate planning to carry out their activities to reduce costs while maintaining health care quality and patient satisfaction. According to previous studies, the present study shows that most studies have examined the problem in terms of logistical costs and have used lower multi-criteria decision-making (MCDM) methods. This has driven our efforts to simultaneously consider the financial, human resource, time and environmental pollutants, and problem-solving through MCDM methods. In this research, a mixed-integer linear programming (MILP) model with three objectives in both definite and indefinite states is presented to optimize the basic HHC processes. In the first objective function, an attempt is made to minimize costs. On the other hand, due to the high importance of environmental pollutants, the second objective function minimizes the greenhouse gas (GHG) emitted by ambulances. Furthermore, for the first time, a new objective function is presented to minimize the congestion of laboratory samples in HHCCs with the M/M/C queueing system. Then, a hybrid model based on the expected value (EV) and fuzzy chance-constrained programming (FCCP) is presented to bring the problem closer to reality. Finally, this mathematical model is solved using goal attainment (GA) and LP-metric methods ranked by a simple additive weighting (SAW) method.

1. Introduction

Home health care (HHC) is a set of services and care provided by nurses and the elderly in their homes. On the other hand, laboratory samples are returned to HHC centers (HHCCs) for processing to rehabilitate the disabled, elderly, and patients to independently perform the daily activities needed for life in their homes and to maintain an acceptable quality of life in the long run (Sinder, 2004). Today, of course, HHCCs can offer more sophisticated therapies that in the past were only available in hospitals, such as geriatric care, chemotherapy, and rehabilitation to prevent stress and occupancy of hospital beds. However, these experiences are still not widespread and are remained localized and separate. Also, HHCCs usually have fragile economic models and depend on the characteristics of the area covered by the HHCC (e.g., kind of population, kind of demand, and proximity to

hospitals). Therefore, creating a sustainable economy with such facilities is very difficult, and significant obstacles still need to be removed. However, patient satisfaction is usually higher in-home treatment than in-hospital treatment (Rodríguez-Verjan et al., 2018).

Since the cost of transportation has become one of the most important costs in the activities of companies, optimizing the ambulance routing problem in the HHC system (HHCS) has become very important (Shi et al., 2017a; Shi et al., 2017b). Based on the United States Environmental Protection Agency GHG sources' general reports, GHG sources are divided into five parts. The calculated GHG emission rate for transportation is estimated at 27%. Transportation is recognized as the main source of pollution in the entire supply chain system and activities related to HHC companies (Zhang et al., 2014). Note that the transportation system's greenhouse gas (GHG) emission rate is vital in the gas emission rate. This motivates this research to discourse the effect of GHG

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rates on the HHCS (Braekers et al., 2016).

On the one hand, increasing the costs of life in advanced countries and more desire to use private care services (e.g., HHC), demand in-HHC has increased, especially in the last decade (Fikar and Hirsch, 2017). In these countries, the population aged 60 and over has increased from 17% in 1980 to 22% in 2004. Also, this factor is likely to increase to 32% in 2030. On the other hand, according to the estimates, an HHC company performs various logistical activities daily, such as sending medicine or medical supplies from HHCCs to patients and delivering laboratory samples and medical waste from the patient and delivering them to medical HHCCs, and also meeting patients according to a schedule. Therefore, due to the increasing demand, if there is no proper planning, there is a possibility of congestion in HHCCs.

On the other hand, for some patients, the time factor is very sensitive. For example, a delay in insulin injection has severe consequences for the patient. For this reason, it is essential to pay attention to the time constraints of service and its uncertainty.

This research aims to present a new mathematical model to cover the gaps in previous studies, such as minimizing the costs associated with HHCCs, such as building HHCCs with possible economic combinations, paying attention to various diseases, and providing services to patients at specific time windows. It is the waiting time for the shipments, taking into account the resulting congestion in the HHCCs. An uncertain model of the problem is presented to bring the problem closer to reality. In the present research, considering the previous research on in-HHC and the gaps between the presented problems and models, an attempt is made to provide a model that comprehensively covers these gaps. The following aspects of the innovation of the research can be mentioned:

- Providing three objective functions in a hybrid fuzzy model for a problem of HHCCs in a green pharmaceutical supply chain (PSC) for the first time.
- Considering specific and different types of patient allocation to HHCCs due to the dependence of the patient's disease type and licenses to serve that disease for the HHCCs.
- Presenting the first research on green ambulance routing and congestion restrictions according to the previous research literature review.
- Covering gaps in previous research models, such as considering the disease type and the queuing system resulting from overcrowding of laboratory samples in-HHCCs, simultaneous picking up and delivery, time window, and the number of human resources simultaneously.
- Solving the model by the LP-metric and goal attainment (GA) methods to balance the cost of the HHC, environmental pollution, and queuing waiting time.
- Ranking the proposed methods by simple additive weighting (SAW) SAW method.

In the continuation of this research, the topics are organized in this way. Section 2 describes essential and related studies in this field. Section 3 presents the model assumptions, the mathematical model of the problem is presented in two definite and indefinite cases. Section 4 offers the solution methods used to solve certain and uncertain models. Section 5 illustrates the numerical results obtained by solving test problems using solution methods. At the end of this section, a ranking of solution methods is performed. At last, conclusions are drawn in Section 8.

2. Literature review

According to the current information, Begur et al. (1997) took into account the HHCS in 1997. A technical decision called the spatial decision support system was used in their study, considering a case study in the United States. Cheng and Rich (1988) proposed a mixed-integer linear programming (MILP) model for the HHCS routing problem. They used a heuristic approach to solve it. Also, Bertels and Fahle (2006)

presented an efficient heuristic approach combining linear programming. The limitations of planning and a set of heuristic approaches developed from previous studies were also seen in their research, especially since their mathematical model was one of the simple problems of ambulance routing.

Due to the novelty of HHC problems, there has been little research in this area in recent years. The purpose of most of these studies is to minimize transportation costs and schedule patient visits by nurses. Their common objective is to optimize the total cost of routing and travel distance (Akjiratikar et al., 2007). From a scientific point of view, basic HHC models are among ambulance routing problems (Shi et al., 2017a; Shi et al., 2017b). In ambulance routing, the goal of a company is to meet the demand of its customers by using several ambulances.

Eveborn et al. (2006) included a decision support system to control decision-making policies following their application to search for an HHCS in Sweden. Eveborn et al. (2009) presented a case study from Sweden on the role of operations research models in strengthening the economic perspectives of HHC institutions.

Rasmussen et al. (2012) presented the problem of scheduling patient appointments by assigning patients to nurses to maximize the services provided. Their mathematical model includes the issue of ambulance routing with time window constraints. They solved their model accurately by the branch-and-price method.

Liu et al. (2013) suggested a different approach based on the tabu search and genetic algorithm methods to solve the HHC routing problem and planning model by adding delivery time and time window. In a similar study, Liu et al. (2014) introduced a combined location-based tabu search method to investigate possible and impossible solutions by solving test problems. In the latest studies, new assumptions have been added to the routing problem and planning model for the HHCS. Manowska et al. (2013) presented a model of daily planning of HHC services by nurses (human resource HHCCs) for patients. Their model includes scheduling unique services for each patient, the skills required by nurses, and the possibility of interdependence between different services needed by each patient. For example, the patient needs two samples of medicine at a definite time, and the other patient needs service by two nurses with different skills. This interdependence between services was first presented in this study. They also considered the time window limit in their model. They solved their proposed mathematical model using both precise and meta-heuristic methods.

A multivariate model supported the HHC routing problem of an Austrian HHC company proposed by Hierman et al. (2015). Another contribution in their research was a two-step method to solving their model. In the first step, the constraint planning method was used as an exact solution method. Also, numerous meta-heuristic methods were used to solve the problem. Their used meta-heuristics included: scatter search, memetic, variable neighborhood search, and simulated annealing meta-heuristic algorithms.

Fikar and Hirsch (2015) used diverse transportation systems (e.g., private cars and public transportation). They also considered walking to the patient, interdependent services, time windows (e.g., compulsory working time), and permitted rest time for nurses. The Austrian Red Cross was studied and solved large and small-scale problems. Based on the difficulties of their problem, a hybrid two-stage meta-heuristic method was used. Yuan et al. (2015) presented a mathematical model for scheduling and routing patient appointments with random service times and the skills required by HHCC nurses. Their mathematical model considered fines for delaying patient visits and solved their problem accurately by a branch-and-price method. Their objective function was to minimize travel costs, fixed costs for nurses, service costs, and fines for delaying patient visits.

While most of the papers in the literature referred to optimizing a single-objective mathematical model, Braekers et al. (2016) first introduced a bi-objective model of HHC routing that includes the total travel distance patient satisfaction. They also showed confusions in their two objective functions. Their main aim was to find a balance between total



Fig. 1. Home health care and pharmaceutical supply chain.

costs and patient satisfaction, in which they used the meta-heuristic neighborhood search method based on location and dynamic planning to achieve their goal. Their proposed model considers nurses with different skills, patients' preferred time to visit, and overtime nurses' cost.

Shi et al. (2017a) introduced indefinite demand using fuzzy logic for the required medicine, and used a hybrid genetic algorithm to represent their criteria, such as the ambulance routing problem with a time window. In another study, Shi et al. (2017b) presented a single-objective model with random travel time and service for ambulance routing and patient allocation to HHCCs. Their model considered routing and scheduling patient appointments according to service time and travel time and was solved by combining a genetic algorithm and a random simulation method. From the obtained results, we can point to the use of developed methods to access better solutions.

Fathollahi-Fard et al. (2018a, 2018b) developed a bi-objective mathematical model of a PSC problem as shown in Fig. 1 and presented a combination of past research. They used four innovative methods of simulated annealing, modified simulated annealing, social spider, modified slap swarm, and a hybrid method of and slap swarm algorithms to solve their NP-hard problem. Their objective function includes the cost and emission rate of pollutants. In their model, they considered the hospital and the laboratory's location. In the following network, it is clear that the nurses' journey starts from the hospital and ends in the laboratory. One of the limitations used is the time window. In another study, Fathollahi-Fard et al. (2018a, 2018b) presented the routing problem and assigning patients. They solved their model with an objective function minimizing the cost using the Lagrange liberation theory method for the first time. In this method, an efficient solution was found between the upper and lower limits of the problem.

Lin et al. (2018) introduced an integrated decision model to consider

both nurse planning and the ambulance routing problem with a timeline. Accordingly, the model in question was an NP-hard problem. They used the hybrid method of the harmonic search algorithm with the genetic algorithm to calculate the optimal solution to their problem.

Rodriguez-Verjan et al. (2018) proposed a design model for the location network of HHCCs and divided their mathematical model into two parts. In the first part, they presented the location problem of HHCCs, and in the second part, they introduced the model of hiring human resources. They divided the whole region into j parts and showed each region's demands according to the type of disease and the time. The only resource used in their case is human resources. The purpose function of their model includes costs related to the opening of HHCCs, allocation of resources related to travel distance, and costs associated with fixed and variable costs of human resources. They asked questions in their research; across the region, HHCCs respond to where they should be located and what resources (i.e., type and number of different resources) are needed to meet demand within a given time horizon. It should be noted that in the proposed model, all requests must be met. Tohidifard et al. (2018) developed a multi-depot HHC routine mathematical model with time windows and fuzzy demands and solved it by particle swarm optimization (PSO) and genetic algorithm (GA). Zahiri et al. (2018) proposed a bi-objective mathematical model for a PSC with product shelf life, substitutability, and quantity discount. They used a robust possibilistic optimization approach to solve their model. Roshan et al. (2019) presented a tri-objective model for a PSC with product perishability and substitutability under crisis and demand uncertainty.

Decerle et al. (2019a) used a memetic algorithm for multi-objective optimization of the HHC problem. Decerle et al. (2019b) optimized the HHC problem with a time window, synchronization, and working time balancing hybrid with the memetic-ant colony optimization algorithm. Dengiz et al. (2019) planned a mathematical model for multiple service

Table 1
Review of the existing literature.

Authors	Year Published	Objective Functions							Constraints		Solution methods			Uncertainty
		Single-Objective	Multi-Objective	Manpower Cost	Waiting Time	Pollutant Emission	Center Establishment Cost	Transportation Cost (Cost, Time, Distance)	Simultaneous Pick-up Delivery	Time Window	Exact	Heuristic	Metaheuristic	Fuzzy
Liu et al.	2013	✓						✓	✓	✓			✓	
Carello and Lanzarone	2014	✓		✓					✓		✓			
Fikar and Hirsch	2014	✓			✓			✓		✓			✓	
Mankowska et al.	2014	✓						✓		✓	✓		✓	
Trautsamwieser and Hirsch	2014	✓			✓			✓		✓	✓		✓	
Bowers et al.	2015	✓						✓	✓			✓		
Hiermann	2015	✓						✓		✓			✓	
Misir et al.	2015	✓		✓				✓		✓		✓		
Yuan et al.	2015	✓						✓			✓			✓
Braekers et al.	2016		✓					✓		✓	✓		✓	
Rodriguez-Verjan et al.	2017	✓		✓			✓	✓			✓			
Shi et al.	2017a	✓						✓		✓		✓	✓	✓
Shi et al.	2017b	✓						✓		✓			✓	
Fathollahi-Fard et al.	2018a		✓			✓		✓		✓		✓	✓	
Fathollahi-Fard et al.	2018b	✓						✓		✓			✓	
Tohidifard et al.	2018	✓						✓		✓	✓		✓	
Fathollahi-Fard et al.	2019		✓			✓		✓					✓	
Moussavi et al.	2019	✓		✓				✓					✓	
Luo et al.	2020	✓				✓		✓			✓		✓	
Shahnejat-Bushehri et al.	2021	✓						✓		✓	✓		✓	
Total		17	3	4	2	3	1	19	3	13	9	4	15	2

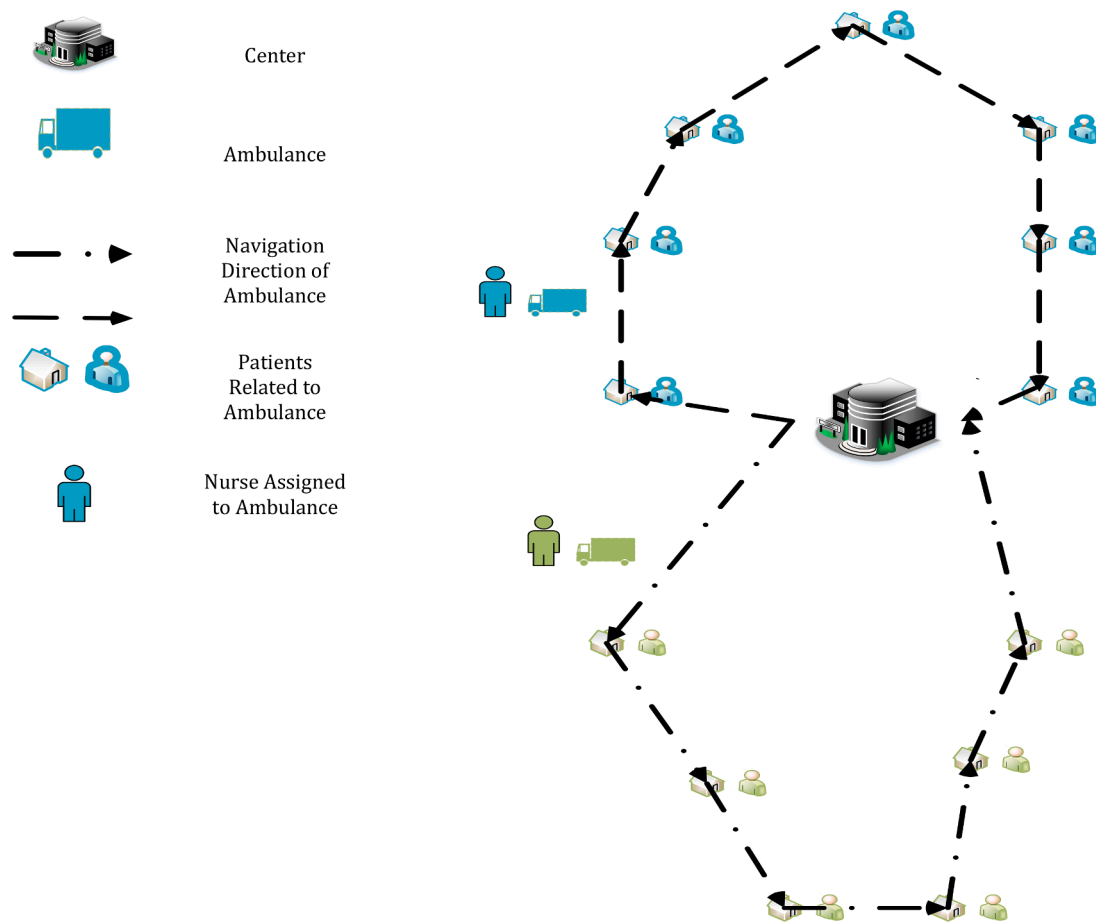


Fig. 2. Structure of the HHC and PSC network proposed in this research.

HHC routing and scheduling problems. Erdem and Koç (2019) analyzed electric vehicles in the HHC routing problem. Fathollahi-Fard et al. (2019) devised a green HHC supply chain and solved it by new modified simulated annealing algorithms. Moussavi et al. (2019) used a meta-heuristic approach to integrate worker assignment and vehicle routing problems with application to HHC scheduling. Grenouilleau et al. (2019) investigated a set partitioning heuristic method for the HHC routing and scheduling problem. Shahnejat-Bushehri et al. (2019) considered HHC routing and scheduling problems with temporal dependencies and perishability with simultaneous pickup and delivery.

Guinet and Dubost (2020) devised a decision-making tool for HHC to discharge conventional hospitals. Liu et al. (2020) planned a HHC problem with continuity of care and a flexible departing way for caregivers. Luo et al. (2020a), Luo et al. (2020b) used an ant colony optimization algorithm (ACO) for a transportation problem in an HHC considering CO₂ emissions. Luo et al. (2020a), Luo et al. (2020b) considered a green routing and scheduling problem in an HHC. Nozir et al. (2020) investigated crew-constrained HHC routing problems with time windows and synchronized visits. Liu et al. (2020) used hybrid meta-heuristics for solving an HHC routing and scheduling problem with time windows, synchronized visits, and lunch breaks. Bahri et al. (2021) investigated the use of electric vehicles in HHC routing problems and analyzed a multi-objective approach under uncertainty. Bazirha et al. (2020) regarded daily scheduling and routing of HHC with multiple availability periods of patients. Chaieb and Ben Sassi (2021) evaluated the HHC scheduling problem with simultaneous pick-up and delivery with a time window using a tabu search method. Dastgoshade et al. (2021) used the Lagrangian relaxation approach for HHC problems.

Ferdowsi et al. (2021) considered a novel routing-scheduling problem for HHC networks. Goodarzian et al. (2021) investigated an

integrated sustainable medical supply chain network during COVID-19. Li et al. (2021) accounted for HHC routing and scheduling problems considering outpatient services. Li et al. (2021) solved a multi-period HHC routing and scheduling problem using an efficient meta-heuristic algorithm. Makboul et al. (2021) investigated a robust approach for the HHC scheduling and routing problem with time windows and synchronization constraints under travel time and service time uncertainty. Fathollahi-Fard et al. (2021) presented a bi-level programming model for the HHC supply chain considering the Stackelberg game theory and outsourcing. They solved their model by three meta-heuristic algorithms: imperialist competitive algorithm, water wave optimization, and salp swarm algorithm. Shakouhi et al. (2021) considered a competitive PSC with a product life cycle, marketing mix strategies, and both Nash and Stackelberg games. Shahnejat-Bushehri et al. (2021) developed HHC routing and scheduling problems considering temporal dependencies and perishability with simultaneous pick-up and delivery. Sisodia and Jindal (2021) used a meta-analysis of industry 4.0 design principles applied in the health sector.

To better illustrate the present study and specify existing research gaps, it is best to categorize them. All studies in the field of HHC indicate that these studies are either to strengthen existing models, solution methods, or are trying to find new models and either offer new solution methods (e.g., Fikar and Hirsch (2015) and Fathollahi-Fard et al. (2018a, 2018b)). This shows that HHC problems can be made more practical and closer to reality by adding more factors. On the other hand, this problem can become an incentive to conduct this research in two parts: presenting a new solution and developing existing optimization models. Recently, researchers considered novel perspectives regarding HHC.

We categorized research works based on attributes regarding the

HHCS problem in Table 1. It is noteworthy that according to this table, the mathematical model presented in this research covers all the objective functions and limitations mentioned in Table 1, which indicates the amount of contribution of this research in HHC problems.

3. Mathematical model

In this study, a tri-objective mathematical model of mixed-integer linear programming with total costs, environmental pollution, and the queuing system are presented simultaneously. Two mathematical models are presented in the definite and fuzzy states. Also, as declared earlier, the key operative decisions of HHC organizations are made daily by numerous supply chains and transportation activities. Medicines and medical supplies are sent to patients from HHCCs. Laboratory samples are then returned from patients' homes to HHCCs. Based on the shown activities, Fig. 2 illustrates the problem.

The previous section presented the types of location, routing, and allocation models mentioned in the subject literature. This section looks at a specific type of tri-objective MILP model presented by ourselves. Our proposed model is of the type of location-allocation-routing problems. This study integrates supply chain problems with location and routing problems and assigns patients to HHCCs. The basic premise is that we consider nodes as potential points for constructing HHCCs. So that is the beginning and the end of all journeys to them. Patients' nodes are assigned to HHCCs according to their type of disease. The nurses deliver the medicines and medical supplies needed by the patients by various ambulances and receive laboratory samples and return medicines from them. These collected shipments are returned to the HHCCs and are delivered to the HHCCs and processed as required. A tri-objective location-allocation-routing model is presented.

3.1. Queue systems

A queuing system can be used as customers or goods that have arrived at using the service or receiving the service, waiting to use the service if it is not immediately available. If they are waiting to use the service, leave the system after service. Queue processes have the following characteristics:

- Customer entry pattern.
- The service model of service providers.
- Queue order.
- System capacity.
- Number of service channels.
- Number of service steps.

It should be noted that the queue model created in each HHCC in the proposed model follows the $M/M/C$ queue model, whose features are presented. In this $M/M/C$ queue model, the queue system has a C server. The service rate of each of these C servers is equal to μ and is independent of the system status (i.e., the number of people in the system). The customer login rate is λ and independent of system status. In this system, there are two modes according to the limited number of customers, so that if the number of customers (n) in the system is less than the number of customers, the customer exit rate will be equal to μ_n , on the other hand, if the number of customers is more than service providers. The customer exit rate will be μ_c , and this is due to the time between the two performed occurrences, which follows the exponential distribution.

The parameters in the queue model are as follows:

λ_n	Customer entry rate when n people are in the system
μ_n	Customer service rate when n people are in the system
p_0	Probability of zero people in the system
p_n	Probability of n people in the system
l_q	Average queue length (i.e., the number of people in the queue).
w_q	Average waiting time for customers in the queue.

The condition of a steady state is $\lambda/(\mu \times c) < 1$. It means that the steady-state condition and the absence of congestion and queuing are that the average input rate to the system should be less than the average potential service rate of the system.

$$\lambda_n = \lambda n = 0, 1, 2, \dots \quad (1)$$

$$\mu_n = \begin{cases} n \cdot \mu & 1 \leq n \leq c \\ c \cdot \mu & c \leq n \end{cases} \quad (2)$$

$$r = \frac{\lambda}{\mu} \quad (3)$$

$$p_0 = \left[\sum_{n=0}^{c-1} \frac{r^n}{n!} + \frac{c \cdot r^c}{c! \cdot (c-r)} \right]^{-1} \quad (1 \leq n \leq c) \quad (4)$$

$$\Rightarrow p_n = \begin{cases} \frac{\lambda^n}{n! \mu^n} p_0 & (1 \leq n \leq c) \\ \frac{\lambda^n}{c^{n-c} c!} p_0 & (n > c) \end{cases} \quad (5)$$

The mean time customers are in queue is as follows:

$$\Rightarrow w_q = \frac{l_q}{\lambda} = \left[\frac{(\lambda/\mu)^c \cdot \mu}{(c-1)! \cdot (c\mu - \lambda)^2} \right]^{-1} \cdot p_0 \quad (6)$$

3.2. Assumptions

The assumptions of the proposed definitive mathematical model are as follows:

- The proposed model includes many patients with known types of diseases, a smaller number of HHCCs with their nurses, and different types of ambulance systems.
- Each nurse can use different types of ambulance systems to visit patients.
- There is no connection between HHCCs.
- Potential locations for the construction of the HHCCs are pre-determined, and a decision should be made for their construction.
- HHCCs with unlimited capacity are provided.
- For each nurse from each HHCC, the beginning and end of the journey is the relevant HHCC.
- The service time of each nurse to each patient is predetermined according to the type of illness and is calculated by the HHCC.
- There is a time window for nurses to be available to serve patients regularly.
- For the total distance traveled, according to the type of transportation system, the maximum amount allowed by the HHCC is considered. If it not complied with, the amount of violation of this amount of the system will include a fine.
- The inventory of each ambulance includes a nurse, medicines, and services for delivery to patients and laboratory samples, expired and expired medicines, and samples collected from patients.
- The amount of cargo sent from the HHCCs to the patients is already known.
- The amount of cargo collected from each patient is known in advance
- All requests must be responded to. This means that all patients must be assigned to one HHCC.
- For visiting patients by nurses, different types of ambulance systems are considered considering various capacities and transportation costs and the emission rate of GHG related to each type of ambulance system.
- Each type of illness has its workload for nurses.

- Any illness requires a license to be serviced. There must be a fixed fee for providing the service conditions and the minimum number of nurses needed to visit them to obtain a permit.
- One of the resources mentioned in the issue is human resources or the nurse, which has limited capacity. In the above problem, each nurse has the capacity of workload, and all nurses' capacity is considered the same.
- The costs of the system are related to transportation, allocating the patient to the HHCC, receiving a permit, establishing the HHCC, and hiring and firing nurses, which are calculated in the model.
- The model of the queuing system of each $M/M/C$ HHCC is considered. The shipments enter the system with a Poisson process at a specific rate. According to the exponential distribution with the considered rate, the service time to the shipments by all servers is the same.

3.3. Indices

$i, j, s \in \{1, 2, \dots, IT\}$	Patient points
$h \in \{1, 2, \dots, HT\}$	Potential points of the HHCC
$k \in \{1, 2, \dots, KT\}$	Ambulance system type
$p \in \{1, 2, \dots, PT\}$	Disease type
$n \in \{1, 2, \dots, NT\}$	Nurse

3.4. Parameters

The following parameters are associated with the considered PSCM and the queues formed back of the HHCCs.

FC_h	Fixed cost of establishing HHCC in place h
γ_p	The cost of obtaining a permit to serve a disease type p
CS	Cost of one demand allocation (patient) unit per unit distance
CC_k	CO ₂ emission rate per unit distance for ambulance system type k
TC_k	Transportation cost per unit distance for ambulance system type k
AD	Amount of fines for nurses and HHCCs for exceeding the allowable distance intended to adapt the model to company policies
A	Cost (salary) of the nurse
A^+	Fixed cost (hiring) increase of one nurse unit
A^-	Fixed cost (firing) reduction of one nurse unit
NO_h	Number of primary care nurses in HHCC h
XP_i	Coordinates of patient node i (horizontal)
YP_i	Coordinates of patient node i (vertical)
XH_h	Coordinates of HHCC node h (horizontal)
YH_h	Coordinates of HHCC node h (vertical)
$DPP_{ij} = \sqrt{(XL_i - XL_j)^2 + (YL_i - YL_j)^2}$	Distance between two nodes i and j (patients)
$DPH_{ih} = \sqrt{(XL_i - XH_h)^2 + (YL_i - YL_h)^2}$	Distance between patient node i and HHCC node h (patients and HHCC)
τ_p	Workload required of each nurse to meet a unit of demand for disease type p
ε	Workload of each nurse is associated with the distance between the HHCC and the patient
δ_p	Demand for disease type p (equal to the total number of patients with disease type p)
CN	Maximum workload capacity of each nurse
η_p	Minimum number of nurses required to serve disease type p to obtain a permit
$TT_{ij} = \frac{DPP_{ij}}{\sum_i \sum_j DPP_{ij}} \times 3600$	Travel time between nodes i and j (patients)
ST_i	Duration of the nurse service to the i patient
M	A very big number
ET_i	Earliest time allowed to serve the i patient
LT_i	Latest time allowed to serve the i patient

(continued on next column)

(continued)

$MLR_{h,n,k}$	Maximum length of the route intended for the nurse n from HHCC h by ambulance system type k that can be navigated
$PD_{p,i}$	Matrix of patients and diseases (in this matrix, 1 if patient i has p disease; 0, otherwise)
CD_i	Patient i delivery demand
CP_i	Patient i pickup demand
CAP_k	Capacity of the ambulance system type k
μ_h	Service rate of HHCC h
NS_h	Number of servers in HHCC h

3.5. Decision variables

This section is considered into two sub-sections.

- Variables associated with the structure of PSCM are considered.
- Variables associated with the queues formed the back of HHCCs.

3.5.1. PSCM variables

y_h	1 if HHCC h is opened; 0, otherwise
$x_{i,j,k}^{h,n}$	1 if from HHCC h nurse n meets patient i before patient j by ambulance system type k ; 0, otherwise
$zz_{i,h}$	1 if patient i is assigned to HHCC h ; 0, otherwise
$b_{h,p}$	1 if HHCC h is permitted to handle disease type p ; 0, otherwise
$q_{p,h}$	1 if disease type p is handled by HHCC h ; 0, otherwise
$a_{k,n,h}$	1 if ambulance type k is assigned to nurse n from HHCC h ; 0, otherwise
$\beta_{n,h}$	1 if nurse n is assigned to HHCC h ; 0, otherwise
nn_h	Number of nurses in HHCC h
nn_h^+	Increased number of nurses in HHCC h
nn_h^-	Decreased number of nurses in HHCC h
$sd_{n,h,k}$	Sum of the distances traveled by nurse n from HHCC h using the ambulance system type k
$ss_{i,n,h}$	Determining the start time of service to patient i by nurse n from HHCC h (if nurse n does not meet patient i , this variable is equal to zero)
$cv_{i,j}$	Amount of medical cargo delivered by the ambulance along routes i to j
$lv_{i,j}$	Amount of medical cargo picked up by the ambulance along routes i to j

3.5.2. Variables associated with the queues

λ_h	Rates of shipments entering HHCC h
p_{0h}	Probability of zero cargo in HHCC h
wq_h	Average waiting time for shipments in line at HHCC h

3.6. Objective functions

To this, we elucidate three objective functions taken into account in our mathematical model. We classify them so that they appear in our mathematical model as shown below:

3.6.1. Minimizing the total cost as the first objective function:

The first objective function tries to minimize the total system costs and consists of six terms. The first term consists of the cost of establishing HHCC h . The second term shows the cost of obtaining an HHCC h permit to service on-demand disease type p . The third term calculates the cost of transporting the system. The fourth part consists of two parts: variable cost (i.e., salary) and fixed cost (i.e., hiring and firing) of the nurse in the HHCC, respectively. The first part calculates the salary paid to the total number of nurses in each HHCC h . The second part also calculates the related costs in case of hiring a nurse or associated costs in case of dismissal of a nurse.

The fifth term calculates the HHCC penalty for each nurse crossing the maximum distance allowed by the transportation system. In this

way, if the total distance traveled by each nurse exceeds the allowable limit by a known transportation system, a penalty will be calculated for the HHCC. This section is intended to adapt the mathematical model of the problem to the policies of the system. The final term is the cost of allocating the patient to the HHCC according to the distance between the HHCC and the patient.

$$nn_h = (N0_h \cdot y_h) + nn_h^+ - nn_h^- \forall h \quad (11)$$

$$nn_h^+ \leq M \cdot y_h \forall h \quad (12)$$

$$nn_h^- \leq M \cdot y_h \forall h \quad (13)$$

$$Z_1 = \text{Min} \left(\sum_h FC_h \cdot y_h + \sum_h \sum_p \gamma_p \cdot b_{h,p} + \sum_h \sum_k \sum_i \sum_j \sum_n DPP_{ij} \cdot TC_k \cdot x_{ij,k}^{h,n} + \sum_h (A \cdot nn_h + A^+ \cdot nn_h^+ + A^- \cdot nn_h^-) + \sum_h \sum_n \sum_k AD \cdot TC_k \cdot sd_{n,h,k} + \sum_i \sum_h CS \cdot DPH_{i,h} \cdot zz_{i,h} \right) \quad (7)$$

3.6.2. Minimizing the emitted CO₂ through the supply chain system:

The second objective function is formulated as Expression (8). This objective function aims to minimize the total emitted GHG (i.e., CO₂) by all considered ambulances.

$$Z_2 = \text{Min} \left(\sum_k \sum_i \sum_j \sum_h \sum_n CC_k \cdot DPP_{ij} \cdot x_{ij,k}^{h,n} \right) \quad (8)$$

3.6.3. Minimizing the total mean waiting time for handling medical cargos, nurses, and related ambulances in the proposed system:

The objective function (9) aims to minimize the mean waiting time in the formed queue before HHCC h to serve the related medical cargo, nurses, and related ambulances.

$$Z_3 = \text{Min} \left(\sum_h wq_h \right) \quad (9)$$

3.7. Proposed tri-objective PSC mathematical model

$$\eta_p \cdot b_{p,h} \leq nn_h \forall p, h \quad (14)$$

$$q_{p,h} \leq b_{p,h} \forall p, h \quad (15)$$

$$b_{p,h} \leq y_h \forall p, h \quad (16)$$

$$\sum_h zz_{i,h} = 1 \forall i \quad (17)$$

$$\sum_h \sum_n \sum_k \sum_j x_{ij,k}^{h,n} = 1 \forall i \quad (18)$$

$$\sum_j \sum_k x_{o,j,k}^{h,n} = \beta_{n,h} \forall n, h \quad (19)$$

$$\sum_i \sum_k x_{i,o,k}^{h,n} = \beta_{n,h} \forall n, h \quad (20)$$

$$\sum_i x_{i,s,k}^{h,n} - \sum_j x_{s,j,k}^{h,n} = 0 \forall h, n, s, k \quad (21)$$

$$\sum_i \sum_j \sum_k x_{ij,k}^{h,n} \leq M \cdot \beta_{n,h} \forall h, n \quad (22)$$

$$Z_1 = \text{Min} \left(\sum_h FC_h \cdot y_h + \sum_h \sum_p \gamma_p \cdot b_{h,p} + \sum_h \sum_k \sum_i \sum_j \sum_n DPP_{ij} \cdot TC_k \cdot x_{ij,k}^{h,n} + \sum_h (A \cdot nn_h + A^+ \cdot nn_h^+ + A^- \cdot nn_h^-) + \sum_h \sum_n \sum_k AD \cdot TC_k \cdot sd_{n,h,k} + \sum_i \sum_h CS \cdot DPH_{i,h} \cdot zz_{i,h} \right)$$

$$Z_2 = \text{Min} \left(\sum_k \sum_i \sum_j \sum_h \sum_n CC_k \cdot DPP_{ij} \cdot x_{ij,k}^{h,n} \right)$$

$$Z_3 = \text{Min} \left(\sum_h wq_h \right)$$

s.t.

$$\sum_p \sum_i (\tau_p + \varepsilon \cdot DPH_{i,h}) \cdot \delta_p \cdot q_{p,h} \leq CN \cdot nn_h \forall h \quad (10)$$

$$\sum_n \beta_{n,h} = nn_h \forall h \quad (23)$$

$$\sum_k \alpha_{k,n,h} = \beta_{n,h} \forall n, h \quad (24)$$

$$\sum_i \sum_j x_{ij,k}^{h,n} \leq M \cdot \alpha_{k,n,h} \forall h, n, k \quad (25)$$

$$ss_{i,n,h} + TT_{ij} + ST_i - M \cdot (1 - x_{ij,k}^{h,n}) \leq ss_{j,n,h} \forall i, j, n, h, k \quad (26)$$

$$ET_i \leq ss_{i,n,h} \leq LT_i \forall i, n, h \quad (27)$$

$$sd_{n,h,k} = \text{Max}(0, (\sum_i \sum_j \text{DPP}_{ij} \cdot x_{ij,k}^{h,n}) - \text{MLRI}_{h,n,k}) \forall n, h, k \quad (28)$$

$$zz_{i,h} \leq y_h \forall i, h \quad (29)$$

$$\sum_n \sum_k x_{ij,k}^{h,n} \leq M \cdot zz_{i,h} \forall i, j, h \quad (30)$$

$$zz_{i,h} \leq \sum_p \text{PD}_{p,i} \cdot q_{p,h} \forall i, h \quad (31)$$

$$\sum_j cv_{i,j} - \sum_j cv_{j,i} = \text{CD}_i \forall i \quad (32)$$

$$\sum_j lv_{i,j} - \sum_j lv_{j,i} = \text{CP}_i \forall i \quad (33)$$

$$cv_{i,j} + lv_{i,j} \leq \sum_k \sum_h \sum_n \text{CAP}_k \cdot x_{ij,k}^{h,n} \forall i, j \quad (34)$$

$$\lambda_h = \sum_i \text{CD}_i \cdot zz_{i,h} \forall h \quad (35)$$

$$p_{0h} = \left[\sum_{n=0}^{NS_h-1} \frac{\left(\frac{\lambda_h}{NS_h}\right)^n}{n!} + \frac{NS_h \cdot \left(\frac{\lambda_h}{NS_h}\right)^{NS_h}}{NS_h \cdot \left(NS_h - \left(\frac{\lambda_h}{NS_h}\right)\right)!} \right]^{-1} \forall h \quad (36)$$

$$wq_h = \left[\frac{\left(\frac{\lambda_h}{NS_h}\right)^{NS_h} \cdot \mu_h}{(NS_h - 1)! \cdot (NS_h \cdot \mu_h - \lambda_h)} \right] \cdot p_{0h} \forall h \quad (37)$$

$$\lambda_h, p_{0h}, wq_h, sd_{n,h,k}, ss_{i,n,h}, cv_{i,j}, lv_{i,j} \geq 0 \forall h, n, k, i, j \quad (38)$$

$$y_h, x_{ij,k}^{h,n}, zz_{i,h}, b_{h,p}, q_{p,h}, \alpha_{k,n,h}, \beta_{n,h} \in \{0, 1\} \quad (39)$$

Constraint (10) indicates that nurses' workload in HHCC h is related to the demand for p -type disease and the distance between patients and HHCC h provided that disease type p is served by HHCC h . Constraint (11) indicates the number of nurses in HHCC h after hiring or firing plus the number of primary nurses in the HHCC. Constraints (12) and (13) ensure that increased and decreased nurses in HHCC h get value if HHCC h is established previously. Constraint (14) guarantees the minimum number of nurses required at HHCC h to serve p -type disease demand if the permit provides disease type p demand. Constraint (15) ensures that when the disease type p demand is allocated to HHCC h , a service permit is obtained for the disease type p demand. Constraint (16) indicates that if HHCC h is not established, it will not receive a service permit. Constraint (17) indicates that any patient can be assigned to one HHCC (i.e., limit patient allocation to HHCC). Constraint (18) emphasizes that each patient should be seen only once. Constraint (19) emphasizes that each nurse n should start her journey from HHCC h provided that nurse n is assigned to HHCC h . Constraint (20) indicates that each nurse n returns to HHCC h after visiting all patients, provided that nurse n is assigned to HHCC h . Constraint (21) indicates that nurse n leaves after seeing each patient. Constraint (22) ensures that nurse n can visit patients from HHCC h when nurse n is assigned to HHCC h . Constraint (23) shows the total number of nurses assigned to HHCC h , equal to the number of nurses in HHCC h . Constraint (24) indicates that an ambulance from HHCC h can be assigned if nurse n is assigned HHCC h .

Constraint (25) indicates that nurse n can meet patient j from HHCC h by ambulance type k after patient i when ambulance type k is assigned to nurse n from HHCC h . Constraints (26) and (27) are time window restrictions. In other words, Constraint (26) ensures that nurse n from HHCC h cannot start serving patient i earlier than the travel time between patient i and j , plus time to serve patient i . All of these constraints

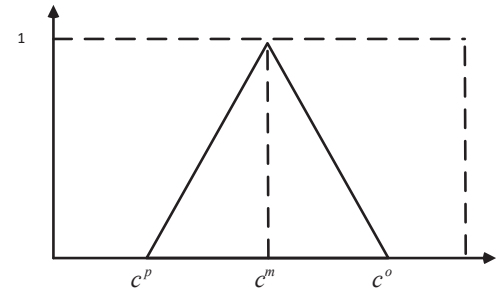


Fig. 3. Triangular fuzzy membership function.

apply when patient j meets after patient i . Constraint (27) Indicates the time window in the issue (i.e., minimum and maximum times allowed to start serving patient i). Constraint (28) shows the total distance traveled by nurse n from the HHCC h by the ambulance system type k . If the total distance traveled exceeds the maximum allowable distance in the problem, the variable related to the sum of the distances traveled by nurse n from HHCC h using the ambulance system type k is obtained value. Otherwise, the value of the related variable is zero, and no penalty is imposed on the system. Constraint (29) ensures that patient i can be assigned to HHCC h when HHCC h is established. Constraint (30) ensures that nurse n from HHCC h by ambulance system type k can meet patient j after patient i when patient j is assigned to HHCC h . Constraint (31) indicates that patient i with type p can be assigned to HHCC h if p is assigned to HHCC h .

Constraints (32) and (33) indicate the limit of simultaneous pickup and delivery. In other words, Constraint (32) determines the exact amount of medical cargo delivered in the ambulance type k along the route i to j . Constraint (33) indicates the exact amount of load carried by the ambulance type k along route i to route j . Constraint (34) suggests the limitation of the capacity of ambulance system type k so that the sum of the delivered and picked-up loads along the route should not exceed the capacity of ambulance system type k . Constraint (35) calculates the rate of shipments entering HHCC h equal to the total number of shipments (i.e., including laboratory samples and returned medicines) received by patients at HHCC h . Constraint (36) calculates the probability of zero medical cargo in HHCC h . Constraint (37) yields the average waiting time elapsed in HHCC h to be served nurses, medical cargo, and related ambulance accordingly. Constraint (38) and (39) defined the positive and binary values, respectively.

3.8. Linearization of the model

A maximization statement $sd_{n,h,k} = \text{Max}(0, (\sum_i \sum_j \text{DPP}_{ij} \cdot x_{ij,k}^{h,n}) - \text{MLRI}_{h,n,k})$ in Constraint (28) should be linearized. The linearization steps are as follows:

$$sd_{n,h,k} \geq 0 \forall n, h, k \quad (40)$$

$$sd_{n,h,k} \geq (\sum_i \sum_j \text{DPP}_{ij} \cdot x_{ij,k}^{h,n}) - \text{MLRI}_{h,n,k} \forall n, h, k \quad (41)$$

3.9. Fuzzy uncertainty concept and regarded mathematical model

In this section, we use the method proposed by Jiménez (1996) for the fuzzy model. Various methods have been proposed to crisp the fuzzy mathematical models, in which the uncertainty is considered in this paper. Some examples are the Max-Min Werners convex combinatorial method (Werners, 1987), Jimenez's method (Jiménez, 1996), and the Max-Min method designed by Bellman and Zadeh (1970). Jiménez's method (Jiménez, 1996) is used in this paper because of its simplicity and flexibility compared to the other methods.

3.9.1. Concept of the EV method

If the fuzzy number is triangular in the form $\tilde{c} = (c^p, c^m, c^o)$, its membership function is calculated below (as shown in Fig. 3).

$$\mu_{\tilde{c}}(x) = \begin{cases} f_c(x) = \frac{x - c^p}{c^m - c^p} & \text{if } c^p \leq x \leq c^m \\ 1 & \text{if } x = c^m \\ g_c(x) = \frac{c^o - x}{c^o - c^m} & \text{if } c^m \leq x \leq c^o \\ 0 & \text{if } x \leq c^p \text{ or } x \geq c^o \end{cases} \quad (42)$$

To ensure that inverse functions $f_c^{-1}(x)$ and $g_c^{-1}(x)$ exist, it is assumed that the ascending and descending functions are continuous, respectively. Consider the cut r of a $\tilde{c}$ fuzzy number as follows:

$$EI(\tilde{c}) = [E_1^c, E_2^c] = \left[\int_0^1 f_c^{-1}(x), \int_0^1 g_c^{-1}(x) \right] = \left[\frac{1}{2}(c^p + c^m), \frac{1}{2}(c^m + c^o) \right] \quad (43)$$

The waiting time of a triangular fuzzy number is calculated by:

$$EI(\tilde{c}) = [E_1^c, E_2^c] = \left[\int_0^1 f_c^{-1}(x), \int_0^1 g_c^{-1}(x) \right] = \left[\frac{1}{2}(c^p + c^m), \frac{1}{2}(c^m + c^o) \right] \quad (44)$$

Finally, the expected value for triangular fuzzy numbers is obtained by:

$$EV(\tilde{c}) = \frac{E_1^c + E_2^c}{2} = \frac{c^p + 2c^m + c^o}{4} \quad (45)$$

According to Jimenez's ranking method (Jimenez, 1996) for both fuzzy numbers A and B , the membership degree of A larger than B is as follows:

$$\mu_M(A, B) = \begin{cases} 0 & \text{if } E_2^A - E_2^B < 0 \\ \frac{E_2^A - E_1^B}{E_2^A - E_1^B - (E_1^A - E_2^B)} & \text{if } 0 \in [E_1^A - E_2^2, E_2^A - E_1^1] \\ 0 & \text{if } E_1^A - E_2^B > 0 \end{cases} \quad (46)$$

3.9.2. Hybrid of EV and FCCP methods

As mentioned earlier, due to uncertainty about the declared fuzzy parameters, decision-making in uncertainty conditions will be associated with risk. Therefore, the DM wants to escape the risk as much as possible and reach a confidence level. In such situations, to help the DM, an attempt is made to enter the expected level of confidence of the DM into the model. One of the most effective methods in case of uncertainty is FCCP.

Chance-constrained programming was first introduced and developed as a powerful tool for random decision systems by Charans and Cooper (1959). In this model, the degree of confidence α is introduced by the DM to have an appropriate confidence level, so that all constraints in the case of uncertainty apply at least to the degree of confidence in the relevant constraint.

Suppose x is a decision variable vector and ξ is a fuzzy parameter vector. $f(x, \xi)$ is an objective function including variables and fuzzy parameters. Also, $g_i(x, \xi)$ with $i = 1, 2, \dots, p$ is a fuzzy uncertain constraint. Since $g_i(x, \xi) \leq 0$ is a fuzzy constraint, a degree of confidence α is defined for it as follows:

$$Cr\{g_i(x, \xi) \leq 0\} \geq \alpha, i = 1, \dots, p \quad (47)$$

To transform the fuzzy uncertainty model presented in this research, we use the hybrid model of the EV and FCCP, which was first proposed by Pishvae et al. (2012). In this method, the confidence level of constraints is controlled without increasing complexity.

The expected value model is used for the fuzzy objective functions, and the FCCP model is used to handle the fuzzy constraints. Considering the basic fuzzy programming model as follows:

$$\begin{aligned} & \max f(X, \xi) \\ & \text{s.t.} \\ & g_i(X, \xi) \leq 0 \quad \forall i = 1, \dots, p \end{aligned} \quad (48)$$

Now the proposed combinatorial model for converting the above fuzzy model to its equivalent as discussed above is regarded below:

$$\begin{aligned} & \max E[f(X, \xi)] \\ & \text{s.t.} \\ & Cr\{g_i(X, \xi) \leq 0\} \geq \alpha_i \quad \forall i = 1, \dots, p \end{aligned} \quad (49)$$

According to Ghodrattnama et al. (2012), critical values can be converted the above model based on confidence. If all the fuzzy parameters of the model are assumed to be triangular numbers (for example $\xi = (r_1, r_2, r_3)$), their critical values considering the related confidence level $\alpha \in [0, 1]$ can be computed by:

$$\xi_{\sup}(\alpha) = \sup\{r | Cr\{\xi \geq r\} \geq \alpha\} \quad (50)$$

$$\xi_{\inf}(\alpha) = \inf\{r | Cr\{\xi \leq r\} \geq \alpha\} \quad (51)$$

Equations (52) and (53) are associated with the α -optimistic and α -pessimistic values.

$$\xi_{\sup}(\alpha) = \begin{cases} 2\alpha \cdot r_2 + (1 - 2\alpha) \cdot r_3, & \text{if } \alpha \leq 0.5 \\ (2\alpha - 1) \cdot r_1 + (2 - 2\alpha) \cdot r_2, & \text{if } \alpha > 0.5 \end{cases} \quad (52)$$

$$\xi_{\inf}(\alpha) = \begin{cases} (1 - 2\alpha) \cdot r_1 + 2\alpha \cdot r_2, & \text{if } \alpha \leq 0.5 \\ (2 - 2\alpha) \cdot r_2 + (2\alpha - 1) \cdot r_3, & \text{if } \alpha > 0.5 \end{cases} \quad (53)$$

Equations (54)-(55) explain how to calculate α -optimistic and α -pessimistic values given confidence level α , respectively. On the other hand, Liu (2007) showed that:

$$Cr\{\xi \geq k\} \geq \alpha \Leftrightarrow k \leq \xi_{\sup}(\alpha) \quad (54)$$

$$Cr\{\xi \leq k\} \geq \alpha \Leftrightarrow k \geq \xi_{\inf}(\alpha) \quad (55)$$

3.9.3. Fuzzy mathematical model

In this section, a tri-objective MILP model with fuzzy parameters is presented. The purpose of presenting this model is to bring the issue as close as possible to reality. It should be noted that the majority of assumptions of this section are the same as the definitive model with except that the four parameters are regarded as uncertain ones. The fuzzy parameters of the model are as follows:

CO ₂ emission rate per unit distance for ambulance system type k	$\widehat{CC}_k$
Travel time between nodes i and j (i.e., patient)	$\widehat{TT}_{ij}$
Duration of the nurse service to patient i	$\widehat{ST}_i$
Patient i pickup demand	$\widehat{CP}_i$

The fuzzy mathematical model formula is as follows:

$$\begin{aligned} Z_1 = & \min \left(\sum_h FC_h \cdot y_h + \sum_h \sum_p \gamma_p \cdot b_{h,p} + \sum_h \sum_k \sum_i \sum_j \sum_n DPP_{ij} \cdot TC_k \cdot x_{i,j,k}^{h,n} + \right. \\ & \left. + \sum_h (A \cdot nm_h + A^+ \cdot nm_h^+ + A^- \cdot nm_h^-) + \sum_h \sum_n \sum_k AD \cdot TC_k \cdot sd_{n,h,k} + \sum_i \sum_h CS \cdot DPH_{i,h} \cdot z_{i,h} \right) \end{aligned} \quad (56)$$

$$Z_2 = \text{Min} \left(\sum_k \sum_i \sum_j \sum_h \sum_n \widetilde{CC}_k \cdot \text{DPP}_{ij} \cdot x_{ij,k}^{h,n} \right) \quad (57)$$

$$Z_3 = \text{Min} \left(\sum_h wq_h \right) \quad (58)$$

s.t.

$$ss_{i,n,h} + \widetilde{TT}_{ij} + \widetilde{ST}_i - M \cdot \left(1 - x_{ij,k}^{h,n} \right) \leq ss_{j,n,h} \forall i, j, n, h, k \quad (59)$$

$$\sum_j lv_{ij} - \sum_j lv_{ji} = \widetilde{CP}_i \forall i \quad (60)$$

$$\lambda_h = \sum_i \widetilde{CP}_i \cdot zz_{i,h} \forall h \quad (61)$$

Constraints (15)-(25), (27), (29)-(32), (34), (36)-(41).

Considering credibility for related constraints with uncertain parameters, we have the following mathematical expressions:

$$Z_2 = \text{Min} \left(\sum_k \sum_i \sum_j \sum_h \sum_n \left(\frac{CC_k^1 + 2CC_k^2 + CC_k^3}{4} \right) \cdot \text{DPP}_{ij} \cdot x_{ij,k}^{h,n} \right) \quad (69)$$

$$Z_3 = \text{Min} \left(\sum_h wq_h \right) \quad (70)$$

s.t.

$$ss_{i,n,h} + (2 - 2\alpha) \cdot TT_{ij}^2 + (2\alpha - 1) \cdot TT_{ij}^3 + (2 - 2\alpha) \cdot ST_i^2 + (2\alpha - 1) \cdot ST_i^3 - M \cdot \left(1 - x_{ij,k}^{h,n} \right) \leq ss_{j,n,h} \quad \forall i, j, n, h, k \quad (71)$$

$$\sum_j lv_{ij} - \sum_j lv_{ji} \leq \left(2 \cdot \left(\frac{\alpha}{2} \right) - 1 \right) \cdot CP_i^1 + \left(2 - 2 \cdot \left(\frac{\alpha}{2} \right) \right) \cdot CP_i^2 \forall i \quad (72)$$

$$\sum_j lv_{ij} - \sum_j lv_{ji} \leq \left(2 - 2 \cdot \left(\frac{\alpha}{2} \right) \right) \cdot CP_i^2 + \left(2 \cdot \left(\frac{\alpha}{2} \right) - 1 \right) \cdot CP_i^3 \forall i \quad (73)$$

$$\lambda_h \leq \left(2 \cdot \left(\frac{\alpha}{2} \right) - 1 \right) \cdot CP_i^1 + \left(2 - 2 \cdot \left(\frac{\alpha}{2} \right) \right) \cdot CP_i^2 \forall h \quad (74)$$

$$Z_1 = \text{Min} \left(\sum_h FC_h \cdot y_h + \sum_h \sum_p \gamma_p \cdot b_{h,p} + \sum_h \sum_k \sum_i \sum_j \sum_n \text{DPP}_{ij} \cdot TC_k \cdot x_{ij,k}^{h,n} + \sum_h (A \cdot nn_h + A^+ \cdot nn_h^+ + A^- \cdot nn_h^-) + \sum_h \sum_n \sum_k AD \cdot TC_k \cdot sd_{n,h,k} + \sum_i \sum_h CS \cdot DPH_{i,h} \cdot zz_{i,h} \right) \quad (62)$$

$$Z_2 = \text{Min} \left(\sum_k \sum_i \sum_j \sum_h \sum_n \widetilde{CC}_k \cdot \text{DPP}_{ij} \cdot x_{ij,k}^{h,n} \right) \quad (63)$$

$$Z_3 = \text{Min} \left(\sum_h wq_h \right) \quad (64)$$

$$Cr \left\{ ss_{i,n,h} + \widetilde{TT}_{ij} + \widetilde{ST}_i - M \cdot \left(1 - x_{ij,k}^{h,n} \right) \leq ss_{j,n,h} \right\} \geq \alpha \forall i, j, n, h, k \quad (65)$$

$$Cr \left\{ \sum_j lv_{ij} - \sum_j lv_{ji} = \widetilde{CP}_i \right\} \geq \alpha \forall i \quad (66)$$

$$Cr \left\{ \lambda_h = \sum_i \widetilde{CP}_i \cdot zz_{i,h} \right\} \geq \alpha \forall h \quad (67)$$

Constraints (15)-(25), (27), (29)-(32), (34), (36)-(41).

Now associated with the given description, the crisp mathematical model is as follows:

$$\lambda_h \geq \left(2 - 2 \cdot \left(\frac{\alpha}{2} \right) \right) \cdot CP_i^2 + \left(2 \cdot \left(\frac{\alpha}{2} \right) - 1 \right) \cdot CP_i^3 \forall h \quad (75)$$

Constraints (15)-(25), (27), (29)-(32), (34), (36)-(41).

4. Solution methods

In this section, firstly, we briefly discuss multi-objective (MO) optimization formulation in general. The literature about the most important multi-objective decision-making (MODM) approaches is scrutinized, and at last, the mechanism of two solution approaches used to solve our models are investigated.

4.1. Multi-objective (MO) optimization formulation in general

Following is an explanation of a general MO issue requiring the optimization of n objectives:

$$\min \vec{y} = \vec{F}(\vec{X}) = [f_1(\vec{X}), f_2(\vec{X}), \dots, f_n(\vec{X})] \quad (76)$$

s.t.

$$g_j(\vec{X}) \leq 0 \quad j = 1, 2, \dots, M \quad (77)$$

$$Z_1 = \text{Min} \left(\sum_h FC_h \cdot y_h + \sum_h \sum_p \gamma_p \cdot b_{h,p} + \sum_h \sum_k \sum_i \sum_j \sum_n \text{DPP}_{ij} \cdot TC_k \cdot x_{ij,k}^{h,n} + \sum_h (A \cdot nn_h + A^+ \cdot nn_h^+ + A^- \cdot nn_h^-) + \sum_h \sum_n \sum_k AD \cdot TC_k \cdot sd_{n,h,k} + \sum_i \sum_h CS \cdot DPH_{i,h} \cdot zz_{i,h} \right) \quad (68)$$

$$\vec{X} = [X_1, X_2, \dots, X_p] \in \Omega \quad (78)$$

where the objective vector is $\vec{y}$. The constraints are g_j and $\vec{X}$ are a P -dimensional vector expressing the decision variables in solution space Ω . The objective space is the region covered by the objective vectors.

The feasible space is the subspace of the objective vectors that satisfy the constraints. The utopian solution is the one that achieves all of the objectives.

$$\vec{X}_0^* \in \Omega : \forall \vec{X} \in \Omega, f_i(\vec{X}_0^*) \leq f_i(\vec{X}) \quad \forall i = 1, \dots, n \quad (79)$$

The MO problem is reduced to a single-objective (SO) problem when $n = 1$. In this case, the utopian solution is just the global optimum. Even though it cannot be found, it always exists. However, in the more general scenario when $n > 1$, the utopian solution is rare since the individual objective functions $\{f_i\}$ are conflicting. Rather, an unlimited number of solutions known as nondominated solutions (for which no objective can be enhanced without degrading at least one other) represent various compromises or trade-offs between the objectives.

4.2. MODM state of art

Various approaches exist related to the MODM approach. Below, we investigate the related literature as the most important section.

• Fuzzy Logic Method

One of the techniques used to solve multi-objective problems is the use of fuzzy programming. Zimmermann (1978) developed this technique. One of the advantages of the fuzzy programming method is the very good and appropriate solutions to the problem. Disadvantages include the great amount of computation when the objective functions increase. Also, most MODM problems can be used as a multi-objective linear programming model (MOLP) formulation (Lai and Hwang, 1994). Due to the inherent inaccuracy in human judgment, the parameters of the linear programming model may be uncertain, in which case the model is fuzzy multi-objective linear programming (FMOLP). The uncertainty may involve decision makers' goals or about goals and constraints both together fuzzy multi-objective linear goal programming (FMOLGP).

• Constrained Objective Method

The constrained objective method is one of the best-known methods for solving multi-objective optimization problems. In this way, only one of the main objectives is to minimize while the other objectives are reduced to a constraint. This method is suitable for solving problems where detailed information is available. Myers and Carter (1973) first proposed optimizing the original solution along with limiting other solutions. Biles (1974) extended this method to more than two solutions.

I) Strengths of the constrained objective method.

- By using different values of limits, other effective solutions can be obtained.
- This method can be used for non-convex multi-objective optimization problems.

II) Weaknesses of the limiting method.

- The solution to this problem largely depends on the proper choice of limits. The limits may be set in such a way that the problem has no solution.
- As the number of objectives increases, more information is needed from the user about the scope of the objective function.
- Lexicography Method

Multi-objective decision-making by lexicography is one of the multivariate optimization strategies that result from the prioritization concept. This method was developed firstly by Croft et al. (1992). In this method, first, the objectives are ranked according to their degree of importance. Then the optimization initiates by optimizing the most important objective. If the optimal solution is unique in optimizing the first objective function, the problem is over. Otherwise, the second objective function will be optimized. The process continues in order of importance until the problem is completely resolved.

• Ideal Compromise Programming (CP) Method

Zeleni and Yu (1973) presented a compromise program to help the decision-maker. This method is a set of distance-based methods, and the preferred option is the option that has the shortest distance from the ideal solution. The first step in an agreed plan is to determine the ideal solution. This solution is achieved by optimizing each objective individually according to resource constraints. The ideal solution is often unattainable due to the conflict between objectives. When it is impossible to reach the ideal solution, the solutions closer to the ideal solution give optimal elements or agreed solutions (Francisco and Ali, 2006).

• TOPSIS Method

The method minimizes the distance from the ideal option because it is the optimal solution that has the shortest distance from the positive ideal solution and, at the same time, the longest distance from the negative ideal solution (Lai and Hwang, 1994). By using this method, a multi-objective problem can be transformed into a single-objective one.

• Goal Programming Method

Goal programming is one of the most practical linear programming techniques and is classified as a subset of multi-objective programming models. Charans and Coupe (1978) first introduced this programming model. Their purpose in presenting this model was to move toward several conflicting objectives simultaneously. This model can incorporate decision-making ideals directly into the model and solve the model accordingly.

For example, the goal programming min-max model, also known as the Chebyshev method, was introduced by Flavell and Flavell (1976). In this method, the maximum number of all deviations is minimized compared to the sum of all deviations. The min-max method seeks to achieve a balance between different objectives. It is also the only way to reach an optimal solution to the multi-objective planning problem that is not in the critical points of the problem-solving space. Chang (2007) presented for the first time a multi-choice goal programming method. In this method, due to limited resources and opportunities and incomplete information available to achieve a more appropriate solution, the decision-maker considers several goals for his objectives.

• Min-max method

The idea for this method to solve multi-objective problems is taken from game theory. As game theory deals with conflicting situations, this approach is used to interact with conflicting objectives. The min-max method was developed by Osyczka (1981) and Rao (1987). The min-max method to get the optimal solution tries to find the weakest objective or, in other words, the farthest objective from the optimal solution and maximize it.

• Weighting Sum Method

This method is one of the simplest multi-objective optimization methods. In this method, multiple objectives are multiplied by a weight

Table 2
Parameters related to the numerical test problem.

	Problem No.		
	1	2	3
FC_h	$\bar{uniformint}(10000, 15000)$	$\bar{uniformint}(10000, 15000)$	$\bar{uniformint}(10000, 15000)$
γ_p	$\bar{uniformint}(100000, 125000)$	$\bar{uniformint}(100000, 125000)$	$\bar{uniformint}(100000, 125000)$
CS	2	2	2
CC_k	$\bar{uniform}(0.0640, 0.1181)$	$\bar{uniform}(0.0640, 0.1181)$	$\bar{uniform}(0.0640, 0.1181)$
TC_k	$\bar{uniformint}(3, 8)$	$\bar{uniformint}(3, 8)$	$\bar{uniformint}(3, 8)$
AD	1.5	1.5	1.5
A	$\bar{uniformint}(50000, 100000)$	$\bar{uniformint}(50000, 100000)$	$\bar{uniformint}(50000, 100000)$
A^+	$\bar{uniformint}(3000, 6000)$	$\bar{uniformint}(3000, 6000)$	$\bar{uniformint}(3000, 6000)$
A^-	$\bar{uniformint}(3000, 4500)$	$\bar{uniformint}(3000, 4500)$	$\bar{uniformint}(3000, 4500)$
XP_i	$\bar{uniformint}(0, 1000)$	$\bar{uniformint}(0, 1000)$	$\bar{uniformint}(0, 1000)$
YP_i	$\bar{uniformint}(0, 1000)$	$\bar{uniformint}(0, 1000)$	$\bar{uniformint}(0, 1000)$
τ_p	$\bar{uniform}(0.2, 2.5)$	$\bar{uniform}(0.2, 2.5)$	$\bar{uniform}(0.2, 2.5)$
ε	$\bar{uniform}(0, 0.006)$	$\bar{uniform}(0, 0.006)$	$\bar{uniform}(0, 0.006)$
CN	$\bar{uniformint}(150, 190)$	$\bar{uniformint}(150, 190)$	$\bar{uniformint}(150, 190)$
η_p	$\bar{uniformint}(1, 3)$	$\bar{uniformint}(1, 3)$	$\bar{uniformint}(1, 3)$
ST_i	$\bar{uniformint}(0, 10)$	$\bar{uniformint}(0, 10)$	$\bar{uniformint}(0, 10)$
ET_i	$Rand\{5, 10, \dots, 60\}$	$Rand\{5, 10, \dots, 60\}$	$Rand\{5, 10, \dots, 60\}$
LT_i	$Rand\{100, 110, \dots, 300\}$	$Rand\{100, 110, \dots, 300\}$	$Rand\{100, 110, \dots, 300\}$
M	1000000000	1000000000	1000000000
$MLRI_{h,n,k}$	$\bar{uniformint}(2500, 5000)$	$\bar{uniformint}(2500, 5000)$	$\bar{uniformint}(2500, 5000)$
CD_i	$Rand\{5, 10, \dots, 100\}$	$Rand\{5, 10, \dots, 100\}$	$Rand\{5, 10, \dots, 100\}$
CP_i	$\bar{uniformint}(5, 25)$	$\bar{uniformint}(5, 25)$	$\bar{uniformint}(5, 25)$
CAP_k	$Rand\{200, 250, \dots, 500\}$	$Rand\{200, 250, \dots, 500\}$	$Rand\{200, 250, \dots, 500\}$
μ_h	$\bar{uniformint}(6, 8)$	$\bar{uniformint}(8, 12)$	$\bar{uniformint}(6, 8)$
NS_h	$\bar{uniformint}(2, 4)$	$\bar{uniformint}(2, 4)$	$\bar{uniformint}(2, 4)$
Goal 1	689940.688	532660.548	614457.24
Goal 2	384.692	400.143	331.43
Goal 3	0.361	0.149	0.086

set by the decision-maker and converted into an objective (i.e., the weighted average of objectives). It should be noted that the assigned weights indicate the importance of each of the objectives (Miettinen, 1999).

• Epsilon-constraint Method

The epsilon-constraint method is one of the most commonly used methods for solving multi-objective problems (Mavrotas, 2009). This approach in each iteration considers one objective function, and the other objective functions are considered as constraints, and then solves the total model. We use two methods to solve the proposed certain and indeterministic model (i.e., LP-metric and GA).

Following, we introduce these two methods for solving multi-objective mathematical models in this paper regarding numerical examples.

4.3. LP-metric method

Hwang and Masud (1979) first introduced this method. The LP-metric method received more attention for two reasons:

- Too little information than the DM needs.
- It is simple to use in practice.

The LP-metric method is used for solving multi-objective problems, in which n objectives with confliction become an aggregated objective by normalizing the objectives, and non-dominated solutions can be obtained by solving this aggregated objective function. Consider the objective function vector $\vec{F}(\vec{X}) = (f_1(\vec{X}), f_2(\vec{X}), \dots, f_n(\vec{X}))$ and

Table 3
Indices related to the numerical test problem.

Index	Description	Problem No.		
		1	2	3
IT	No. of patients	9	12	16
HT	No. of potential HHCCs	3	3	4
NT	No. of nurses	3	4	4
KT	No. of various ambulance type	2	2	2
PT	No. of various diseases	2	2	2

Table 4
Number of primary care nurses in each HHCC related to the test problems.

$N0_h$	Description	Problems	HHCCs			
			1	2	3	4
			—	—	—	—
	Number of primary care nurses	1	2	1	2	—
	in the HHCC h	2	2	1	2	—
		3	2	2	0	2

Table 5
Number of demand for disease type related to the test problems.

δ_p	Description	Problems	Disease type	
			1	2
			—	—
	Demand for disease type p (equal to the total number of patients with type p disease)	1	6	3
		2	7	5
		3	10	6

Table 6

Disease type of each patient related to the test problem no.1.

Problem no.1	$PD_{p,i}$	Description	Disease	Patient								
				1	2	3	4	5	6	7	8	9
		Disease vs. patient matrix	1	1	1	0	1	0	0	1	1	1
			2	0	0	1	0	1	1	0	0	0

Table 7

Disease type of each patient related to the test problem no.2.

Problem no.2	$PD_{p,i}$	Description	Disease	Patient											
				1	2	3	4	5	6	7	8	9	10	11	12
		Disease vs. patient matrix	1	1	1	0	1	0	1	0	1	1	0	0	1
			2	0	0	1	0	1	0	1	0	0	1	1	0

Table 8

Disease type of each patient related to the test problem no.3.

Problem no.3	$PD_{p,i}$	Description	Disease	Patient																
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
			-		1	1	1	0	1	0	0	1	1	0	0	1	1	0	1	1
		Disease vs. patient matrix		2	0	0	1	0	1	1	0	0	0	1	1	0	0	1	0	0

Table 9

Considered Scenarios and related weights.

No. of scenarios	Weights		
	W_1	W_2	W_3
1	0.8	0.1	0.1
2	0.1	0.8	0.1
3	0.1	0.1	0.8
4	0.33	0.33	0.33
5	0.45	0.45	0.1
6	0.45	0.1	0.45
7	0.1	0.45	0.45
8	0.8	0.1	0.1

ideal vector $\vec{F}^*(\vec{X}) = (f_1^*(\vec{X}), f_2^*(\vec{X}), \dots, f_n^*(\vec{X}))$. This method minimizes the total relative deviations of the objectives from their optimal value and combines the multiple objective functions into one objective. In other words, the LP-metric approach aims to minimize the distance between two points $\vec{F}(\vec{X})$ and $\vec{F}^*(\vec{X})$, which is defined by:

$$\left. \begin{array}{l} \min f_1(\vec{X}) \\ \min f_2(\vec{X}) \\ \vdots \\ \min f_n(\vec{X}) \\ \text{s.t.} \\ \vec{X} \in x \end{array} \right\} \rightarrow \left\{ \begin{array}{l} \min D_p = \left(\sum_{j=1}^n w_j \cdot \left(\frac{f_j(\vec{X}) - f_j^*(\vec{X})}{f_j^*(\vec{X})} \right)^p \right)^{1/p} \\ \text{s.t.} \\ \vec{X} \in x \end{array} \right. \quad (80)$$

where w_j is the weight value of the objective function j and p is equal to the norm of the distance measurement, so that $1 \leq p \leq \infty$ is such that the larger the value of p , the greater the emphasis on the largest deviations. If $p = \infty$, it means that the largest deviation of the available deviations is considered for optimization. The value of p depends on the mental criteria of the decision-maker (DM). The main aim is to minimize the distance function D_p and find the optimal solutions according to the problem constraints. For additional information, please refer to Zeleny (1974), Zeleny (1982), and Zeleny and Cochrane (1973).

Table 10

First OFV versus weights (problem no.1).

No. of scenarios	Weight (w_1)	Objective function (z_1)	
		LP-metric	GA
1	0.8	715561.2	615243.9
2	0.1	1,059,050	614468.3
3	0.1	955469.2	614542.6
4	0.33	1,070,855	614,546
5	0.45	841357.4	614542.6
6	0.45	744961.7	614967.1
7	0.1	1,054,843	614491.8

Table 11

First OFV versus weights (problem no.2).

No. of scenarios	Weight (w_1)	Objective function (z_1)	
		LP-metric	GA
1	0.8	614367.3	529532.4
2	0.1	938646.4	524733.2
3	0.1	674307.5	527241.6
4	0.33	699276.4	530707.8
5	0.45	917,914	532699.7
6	0.45	731264.6	532836.7
7	0.1	917698.4	529669.3

Table 12

First OFV versus weights (problem no.3).

No. of scenarios	Weight (w_1)	Objective function (z_1)	
		LP-metric	GA
1	0.8	1,155,885	689981.2
2	0.1	1,425,849	689803.3
3	0.1	1,578,256	689899.6
4	0.33	1,578,256	689943.7
5	0.45	1,127,171	689953.9
6	0.45	1,578,256	689899.7
7	0.1	1,578,256	689941.8

Table 13
Second OFV versus weights (problem no.1).

No. of scenarios	Weight (w_2)	Objective function (z_2)	
		LP-metric	GA
1	0.1	596.92	429.77
2	0.8	311.43	419.65
3	0.1	633.91	416.83
4	0.33	565.72	420.16
5	0.45	412.32	416.83
6	0.1	472.54	444.73
7	0.45	371.3	486.78

Table 14
Second OFV versus weights (problem no.2).

No. of scenarios	Weight (w_2)	Objective function (z_2)	
		LP-metric	GA
1	0.1	514.33	439.3
2	0.8	401.3	469.71
3	0.1	541.33	449.29
4	0.33	624.97	459.35
5	0.45	404.13	452.51
6	0.1	574.51	450.1
7	0.45	401.3	456.3

Table 15
Second OFV versus weights (problem no.3).

No. of scenarios	Weight (w_1)	Objective function (z_1)	
		LP-metric	GA
1	0.1	680.25	386.58
2	0.8	506.27	408.44
3	0.1	1095.57	381.86
4	0.33	1095.57	379.6
5	0.45	453.95	377.51
6	0.1	1095.57	381.19
7	0.45	1095.57	389.78

4.4. GA method

The approach for attaining goals entails stating a collection of design objectives or goals $\vec{b}^*$ linked to a set of objective functions $\vec{F}(\vec{X})$ where $\vec{X}$ a vector of variables for optimization. The formulation of the problem allows for over- or under-achievement of goals, setting exceedingly optimistic goals without resulting in an unsolvable dilemma. A proportional weighting w_j is also defined as one of the objectives, allowing the DM should include every desirable goal. However, each should be given a weighted ranking.

The following are the objective function and weighting vectors:

$$\vec{b}^* = [b_1^*, b_2^*, \dots, b_n^*] \quad (81)$$

$$\vec{F}(\vec{X}) = [f_1(\vec{X}), f_2(\vec{X}), \dots, f_n(\vec{X})] \quad (82)$$

$$\vec{w} = [w_1, w_2, \dots, w_n] \quad (83)$$

The problem of optimization is as follows:

$$\left. \begin{array}{l} \min f_1(\vec{X}) \\ \min f_2(\vec{X}) \\ \vdots \\ \min f_k(\vec{X}) \end{array} \right\} \rightarrow \left\{ \begin{array}{l} \min Z \\ \text{s.t.} \\ f_j(\vec{X}) - w_j \cdot Z \leq b_j \\ \vec{X} \in x, \\ Z \text{ free in sign} \end{array} \right. \quad (84)$$

Table 16
Third OFV versus weights (problem no.1).

No. of scenarios	Weight (w_3)	Objective function (z_3)	
		LP-metric	GA
1	0.1	0.134	0.58
2	0.1	0.128	0.59
3	0.8	0.064	0.62
4	0.33	0.06	0.61
5	0.1	0.146	0.6
6	0.45	0.113	0.6
7	0.45	0.071	0.61

Table 17
Third OFV versus weights (problem no.2).

No. of scenarios	Weight (w_3)	Objective function (z_3)	
		LP-metric	GA
1	0.1	0.173	3.95
2	0.1	0.173	3.66
3	0.8	0.159	4.76
4	0.33	0.176	3.91
5	0.1	0.173	3.72
6	0.45	0.256	3.95
7	0.45	0.173	3.83

Table 18
Third OFV versus weights (problem no.3).

No. of scenarios	Weight (w_3)	Objective function (z_3)	
		LP-metric	GA
1	0.1	0.55	386.58
2	0.1	0.17	408.44
3	0.8	0.12	381.86
4	0.33	0.12	379.6
5	0.1	0.43	377.51
6	0.45	0.12	381.19
7	0.45	0.12	389.78

If one of the weighting vector's elements (w_i) is zero, the accompanying objective is meant to be "hard", and the solution shall explicitly satisfy the inequality (if possible). It might be anything depending on their inheritance constraining the optimization parameters $\vec{X}$ is required to a predetermined (linear or nonlinear) region enforced by economic or process-related factors.

The GA method is taken into account as the Min-Min type of the problem. Its purpose is to minimize the deviation from the ideal values of objective functions. For additional information, please refer to [Hwang and Masud \(1979\)](#).

5. Computational results

In this section, the certain and uncertain (i.e., fuzzy) mathematical models of the proposed problem are solved using LP-metric and GA methods.

Table 19
Computational time versus scenarios (problem no.1).

No. of scenarios	Computational time	
	LP-metric	GA
1	2176.36	2035.39
2	2568.17	2077.67
3	2217.83	2036.5
4	2421.24	2131.88
5	2176.36	2059.88
6	2025.63	2085.39
7	2026.60	2151.63

Table 20
Computational time versus scenarios (problem no.2).

No. of scenarios	Computational time	
	LP-metric	GA
1	4372.04	5344.06
2	4410.31	7681.5
3	4416.39	5503.86
4	4339.94	7697.40
5	4352.95	7689.06
6	4351.1	7695.7
7	4346.4	7664.26

Table 21
Computational time versus scenarios (problem no.3).

No. of scenarios	Computational time	
	LP-metric	GA
1	6840.17	6310.84
2	6960.28	6422.99
3	6953.6	6037.33
4	6330.41	6272.69
5	6330.61	6740.28
6	6427.45	6769.32
7	6433.75	6767.53

The obtained results are investigated and analyzed accordingly. The behavior of objective functions concerning the different test problems is scrutinized using various scenarios. Because the problem-solving time increases exponentially as the sizes of the problem increase, small and medium scale sizes are provided. To solve the certain model of the problem, three test problems are considered with different sizes. Also, one test problem is regarded to solve the fuzzy uncertain model. All of the parameters are generated randomly. Finally, to solve the tri-objective mathematical model, GAMS software version 25.1.2 with a system with installed memory (RAM) specifications: 4.00 GB and processor: Intel (R) Core (TM) i5-2450 M CPU @ 2.50 GHz is used.

In solving the model, due to the problem's non-linearity, the BARON solver is used to solve it. Also, our computational results are divided into two main classifications, namely, deterministic and in-deterministic models. Following, we interpret our results in detail. For all defined test problems, the analysis of the objective functions and the computational times are performed.

5.1. Deterministic model

The ideal values of the parameters expressed for each test problem are illustrated in Table 2 and the indices used for different test problems are provided in Table 3. Table 4 shows the number of primary care nurses in each HHCC related to the test problems. Table 5 displays the number of demands for disease type related to the test problems. Tables 6–8 illustrate the disease type of each patient related to the test problem no.1 to 3.

To do the computational analysis, Table 9 is regarded to show the scenarios and objective function-related weights. Note that the total weight for each scenario is equal to one. Also, a variety of weights are considered to investigate obtained Pareto solution.

We scrutinize computational results acquired by the LP-metric and GA methods concerning the primary measures: first, second, and third objective function values, and computational time at last.

5.1.1. First objective function

Regarding problem no.1, results obtained by the LP-metric method have bigger vacillations than the GA method. The LP-metric method for big weights has less objective function value but with some exceptions. For example, for weight 0.8 objective function value is at the minimum point. On the other hand, for weight 0.1 objective function value is in the

Table 22
Decision matrix.

Solution methods	Objective function			CPU-time (s)
	Z ₁	Z ₂	Z ₃	
LP-metric	1,045,595	614.1	0.172	4405.12
GA	611411.7	424.12	2.943	4898.3
Weight	0.3	0.3	0.3	0.1

Table 23
Non-scaled weighted decision matrix.

Solution methods	Objective function			CPU-time (s)
	Z ₁	Z ₂	Z ₃	
LP-metric	0.585	0.691	1	1
GA	1	1	0.0584	0.899
Weight	0.3	0.3	0.3	0.1

Table 24
Results obtained using the SAW method.

Solution methods	Acquired score	Rank
LP-metric	0.783	1
GA	0.708	2

worst situation. Concerning problem no.2, the same habit for problem no.1 is observed. GA has a monotonous trend; however, LP-metric has vacillations for weights 0.8, and 0.1 minimum and maximum objective function values are reported. However, some fluctuations are depicted. For example, for weight 0.45, two different objective function values are reported. However, as a whole, the related trend is logical. Related to problem no.3, like other test problems monotonous trend is reported for the GA method. However, the best and worst objective function values are logical for the biggest and smallest weights. Some exceptions are seen through weights.

Note that for all test problems, the GA method outperforms the LP-metric method regarding all weights. Tables 10–12 show numerical results in detail.

5.1.2. Second objective function

Regarding problem no.1, GA has a monotonous trend but with fewer vacillations. For example, 0.8 minimum objective function value is reported for weights, and worse objective function values are observed for fewer weights. The LP-metric method has a logical trend for all weights, meaning for bigger weights, better objective function value and worse objective function value for smallest weights are reported. Note that for some points, the LP-metric method outperforms the GA method. Concerning problem no.2, the same trend is reported for both methods, meaning that the monotonous values trend is observed for the GA method and the LP-metric logical trend. For the bigger weights, the less objective function value and the bigger objective function value for the smaller weight. However, for some points, LP-metric outperforms the GA method.

About problem no.3, a monotonous trend is observed for the GA method, and some vacillations are reported for LP-metric. The better objective function values are reported for bigger weights, the worse and smaller ones. However, some vacillations are observed. Tables 13–15 show numerical results in detail.

5.1.3. Third objective function

Regarding problem no.1, monotonous trends are observed for both methods. However, the LP-metric outperforms the GA method for all weights. However, a logical trend is reported for the LP-metric. Meaning, the bigger the weights yield, the better the objective function value. However, some exceptions are observed. Note that for both methods,

LP-metric and GA

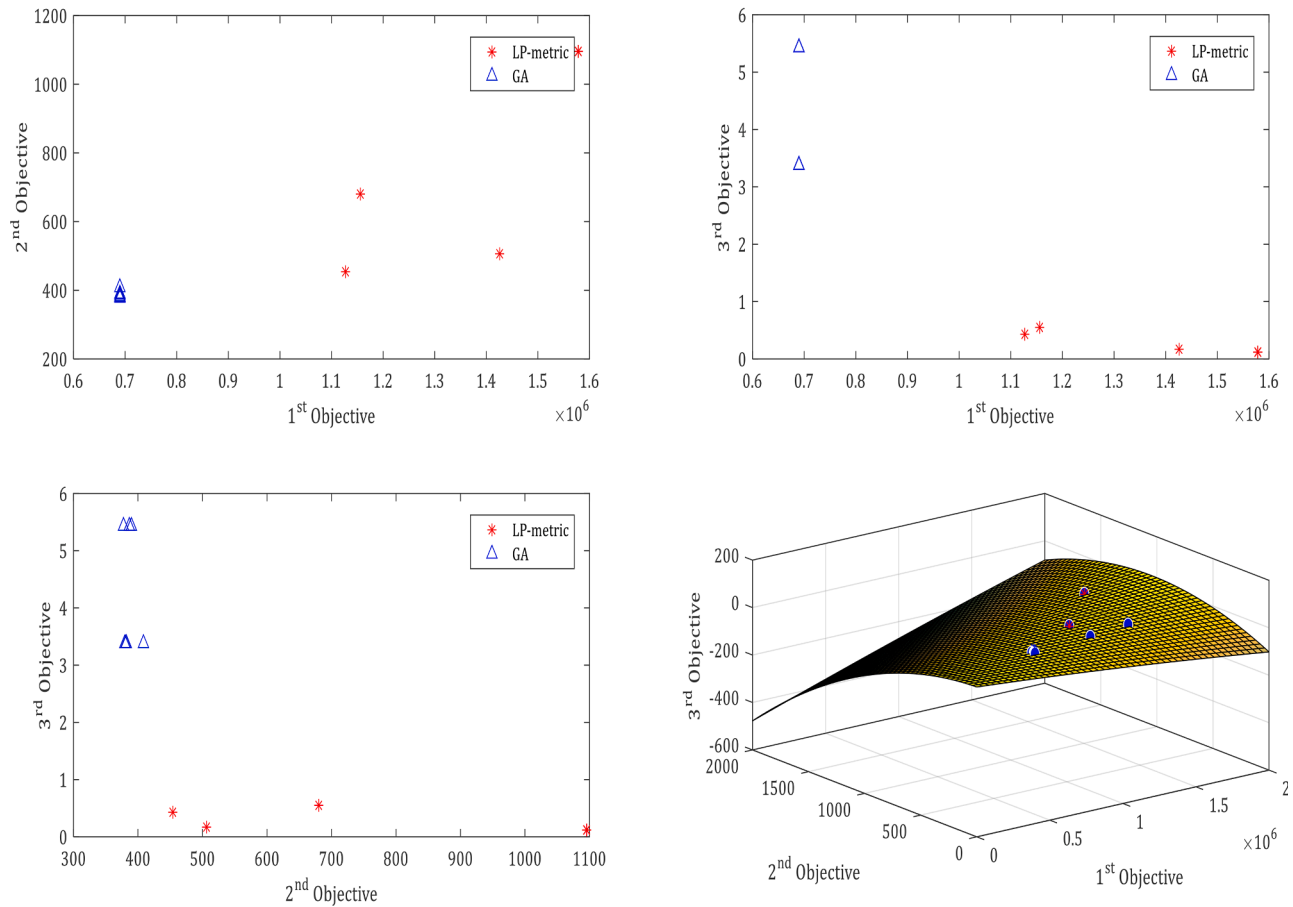


Fig. 4. Non-dominated solutions obtained and formed estimated hyperplane using GA and LP_metric methods.

Table 25

Fuzzy test problem.

Index	Description	Fuzzy problem
IT	No. of patients	9
HT	No. of potential HHCCs	3
NT	No. of nurses	4
KT	No. of various ambulance types	2
PT	No. of various diseases	2

Table 26

Certain parameters related to considered fuzzy test problem.

Parameter	Values	Parameter	Values
Fc_h	$\bar{uniformint}(10000, 15000)$	τ_p	$\bar{uniform}(0.2, 2.5)$
γ_p	$\bar{uniformint}(100000, 125000)$	ε	$\bar{uniform}(0, 0.03)$
CS	2	CN	$\bar{uniformint}(150, 190)$
TC_k	$\bar{uniformint}(3, 8)$	η_p	$\bar{uniformint}(1, 3)$
AD	1.5	EA_i	$\bar{uniformint}(0, 10)$
A	$\bar{uniformint}(50000, 100000)$	LA_i	$Rand\{100, 110, \dots, 300\}$
A^+	$\bar{uniformint}(3000, 6000)$	M	1000000000
A^-	$\bar{uniformint}(3000, 4500)$	$MLRI_{h,n,k}$	$\bar{uniformint}(2500, 5000)$
XP_i	$\bar{uniformint}(0, 1000)$	CD_i	$Rand\{5, 10, \dots, 100\}$
YP_i	$\bar{uniformint}(0, 1000)$	CAP_k	$Rand\{200, 250, \dots, 500\}$
NE_h	$\bar{uniformint}(2, 4)$	μ_h	$\bar{uniformint}(6, 8)$

Table 27

Fuzzy parameters related to considered fuzzy test problem.

Parameter	Values	Parameter	Values
CC_k^1	$\bar{uniform}(0.064, 0.083)$	ST_i^1	$\bar{uniform}(1, 4)$
CC_k^2	$\bar{uniform}(0.083, 0.1015)$	ST_i^2	$\bar{uniform}(4, 8)$
CC_k^3	$\bar{uniform}(0.1015, 0.1181)$	ST_i^3	$\bar{uniform}(8, 12)$
CP_i^1	$\bar{uniform}(5, 10)$	T_{ij}^1	$\bar{uniform}(10, 25)$
CP_i^2	$\bar{uniform}(10, 16)$	T_{ij}^2	$\bar{uniform}(25, 40)$
CP_i^3	$\bar{uniform}(16, 21)$	T_{ij}^3	$\bar{uniform}(40, 55)$

Table 28

Number of primary care nurses in each HHCC related to the fuzzy test problem.

NO_h	Description	HHCC		
		1	2	3
	Number of primary care nurses in the HHCC h (fuzzy problem)	2	1	2

Table 29
Number of demands for disease type related to the fuzzy test problem.

δ_p	Description	Disease Type	
		1	2
	Demand for disease type p (equal to the total number of patients with type p disease)	6	3

there exist some vacillations. Concerning problem no.2, both LP-metric and GA methods have a monotonous habit. LP-metric outperforms GA for all weights. Note that vacillations for LP-metric are very small. However, logical solutions are reported for the LP-metric method (the fewer weights yield, the worse objective function value). Concerning problem no.3, like other problems, the same trend is reported. LP-metric and GA methods have a monotonous trend with small vacillations. The LP-metric method outperforms the GA method. For both methods, the fewer weights yield, the better the objective function value and vice versa. Tables 16–18 show numerical results in detail.

5.1.4. Computational times

Regarding problem no.1, more computational time is reported for LP-metric. However, the GA method has less and uniform computational time through scenarios. The reason may be the solution spaces formed by two methods. Concerning problem no.2, the longer computational time is reported for both methods. The reason is related to the bigger size of the test problems and the resulted bigger solution spaces. LP-metric outperforms the GA method regarding this criterion. As a whole, the bigger vacillations are reported for the GA method.

Associated with problem no.3, there exist no specific trends. However, when problem no increases, the sizes of test problems and solution spaces become bigger. Thus, computation time to report optimum solution becomes bigger. Tables 19–21 illustrate numerical results in detail.

5.1.5. Ranking (LP-metric and GA)

Generally, to define which solution method is superior to the other approach, the SAW method has been used to define the most effective approach in this study. This approach was first presented in 1981 by

Hwang and Yun (Zeleny, 1982). In this approach, the existing data is calculated as a matrix with m rows representing the number of alternatives and n representing the number of criteria. Before beginning the calculations, the weights of each criterion and the kind of criteria should be defined by the DM negatively and positively. In the SAW approach, using the weighted average, we obtain the importance of each alternative and then select the maximum value obtained as the best alternative. The steps of this approach are as below:

Step 1) Using the vector-matrix approach, we remove the scale of the decision matrix (linear unscaling) to compare the alternatives. Considering the following decision matrix:

$$D = \begin{pmatrix} r_{1,1} & \cdots & r_{1,n} \\ \vdots & \ddots & \vdots \\ r_{m,1} & \cdots & r_{m,n} \end{pmatrix} \quad (85)$$

If criterion j is regarded as a positive one (the more value is better), we have:

$$n_{ij} = \frac{r_{ij}}{r_{j,\max}} \quad i = 1, \dots, m \quad ; \quad j = 1, \dots, n \quad (86)$$

If criterion j is regarded as a negative one (the less value is better), we have:

$$n_{ij} = \frac{r_{j,\min}}{r_{ij}} \quad i = 1, \dots, m \quad ; \quad j = 1, \dots, n \quad (87)$$

As a result, we have:

$$N = \begin{pmatrix} n_{1,1} & \cdots & n_{1,n} \\ \vdots & \ddots & \vdots \\ n_{m,1} & \cdots & n_{m,n} \end{pmatrix} \quad (88)$$

Step 2) Multiply each criterion's weight in the column related to that criterion in the unscaled matrix.

$$W = (w_1, w_2, \dots, w_n) \quad (89)$$

$$v_{ij} = n_{ij} \times w_j \quad i = 1, \dots, m \quad ; \quad j = 1, \dots, n \quad (90)$$

$$V = \begin{pmatrix} v_{1,1} & \cdots & v_{1,n} \\ \vdots & \ddots & \vdots \\ v_{m,1} & \cdots & v_{m,n} \end{pmatrix} \quad (91)$$

Table 30
Disease type of each patient related to the fuzzy test problem.

Fuzzy Problem	$PD_{p,i}$	Description	Disease	Patients								
				1	2	3	4	5	6	7	8	9
		Disease vs. patient matrix	1	1	1	0	1	0	0	1	1	1
			2	0	0	1	0	1	1	0	0	0

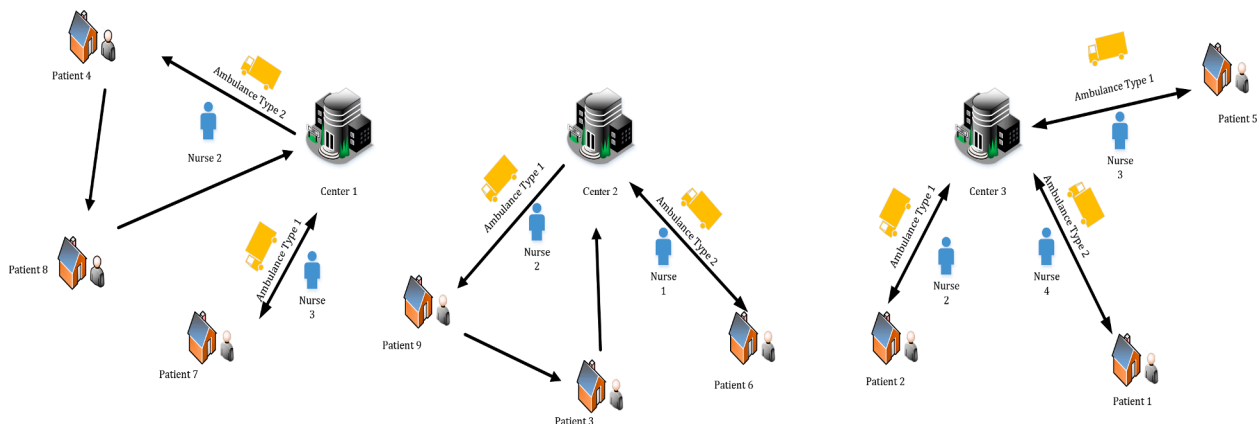


Fig. 5. Network obtained by solving the fuzzy test problem with the confidence level of 0.8.

Table 31
Numerical results obtained related to the considered alternative.

Confidence level	Objective function			CPU-time (s)
	Z_1	Z_2	Z_3	
$\alpha = 0.7$	418452.69	295	2.383	3264.043
$\alpha = 0.8$	418425.595	285.54	2.059	3328.86
$\alpha = 0.9$	439061.29	285.54	0.852	3369.958

Step 3) Obtaining an appropriate alternative regarding the following expression.

$$A^* = \left\{ A_i \mid \max_i \sum_j v_{ij} \right\} \quad (92)$$

To use this method, the mean of objective functions and the computational time are used. As shown above, we use three numerical examples to solve the mathematical model and to calculate the mean of the objective functions and the solution time, the average of the three numerical examples is calculated. The weights are first assigned to each of the objective functions and time. Table 22 shows the average values of the objective functions and the average solution time and weight given to each of these criteria by the DM.

In the next step, the unscaled decision matrix t is presented in Table 23. Then, according to the values obtained and the weights provided by the DM, the ranking table of the solution methods is presented in Table 24. Then, according to the values obtained and the weights provided by the DM, the ranking table of the solution methods is shown in Table 24. Given the rank acquired in this table, it is apparent that the LP-metric approach is better than the GA method.

5.2. Conflicting among the objective functions

To assert the conflict among objective functions. Fig. 4 clarifies these conflicts regarding three viewpoints graphically. This figure compares two solution methods by the Pareto solution obtained using the LP-metric and GA methods.

Note that all objective functions are aimed to be minimized. Also, this figure insists that the conflict, increasing one objective function concerning another objective function, decreases with some exceptions.

5.3. Indeterministic model (fuzzy)

To solve the fuzzy model of the problem, a numerical example is considered, which is solved for different confidence levels to scrutinize the behavior of the objective functions. The test problems size and the corresponding amount of parameters are shown in the following tables. Thus, after determining the values for the fuzzy parameters, we use the hybrid model of the expected value (EV) and fuzzy chance-constrained programming (FCCP) to convert this fuzzy hybrid model to a crisp model. The size of the test problems is depicted in Table 25. The certain parameters are shown in Table 26. The fuzzy parameters are shown in Table 27. Table 28 shows the number of primary care nurses in each HHCC. Table 29 displays the number of demands for disease type. Table 30 illustrates the disease type of each patient. We solve several defined test problems for different α -confidence levels and analyze its objective functions in detail.

Table 31 shows that the values of the objective functions are sensitive to changes in the confidence level. These changes generally worsen the values of the objective functions due to the solved numerical sample with the increasing confidence level. Except for the first objective function, which has a value worse than the confidence level of 0.8 and 0.9 for the confidence level of 0.7, the other objective functions have taken a worse value by increasing the confidence level. On the other hand, we see small changes with the confidence level change in the

second objective function, which can be due to the low CO₂ emission rate. The third objective function is also sensitive to the value of the confidence level and takes the worse values as the confidence level increases.

In Fig. 5 of the network obtained from solving the numerical test, the problem is shown for the confidence level of 0.8. In this example, all three candidate HHCCs are established. The first and second HHCCs do not hire any nurses, and the third HHCC hires a new nurse. The first HHCC only obtain a service permit for type 1 disease. Therefore, as shown in Fig. 5, it serves patients with type 1 disease. However, the second and third HHCCs obtain a permit to serve both types of diseases. Also, none of the nurses exceeds the maximum travel distance allowed. For this reason, no fines are imposed on any HHCC.

6. Conclusions and further research remarks

In this research, modeling and solving a tri-objective supply chain problem in the HHCS was performed by considering the conditions of uncertainty and congestion, and two mathematical models (one in the definite state and one in the fuzzy state) were presented. In the model presented in this research, an attempt has been made to consider the main decisions of HHC issues. The proposed mathematical model is a tri-objective MILP model with a combination of objective functions and constraints first proposed. The first objective function sought to optimize location, transportation, allocation, and human resource costs. Also, the gas emission rate was considered according to the type of transportation system. On the other hand, the congestion resulting from the admission and the initial processes on the laboratory samples in the HHCCs were also addressed in the mathematical model. The model was solved in a period of time using the LP-metric and GA methods. Two different methods were used to solve the problem to provide a combination of efficient answers. Of course, not all the solutions were effective. Then, using the simple weighting method, we ranked between the two methods, which according to the result of the simple weighting method, the LP-metric method gave relatively better results. Then, to get closer to the reality of the fuzzy model, the problem was presented, and in 1 example, a numerical example was solved by considering different levels of reliability. According to the obtained values, with increasing the confidence level of the problem, the values of the objective functions took relatively worse values.

For further research remarks, our suggestions are as follows:

- Due to the complexity of the mathematical model of the problem and the relatively high number of constraints, it is impossible to solve the model in large dimensions by commercial software. Meta-heuristic algorithms can be used to reduce solution time and access to more appropriate solutions to large-scale problems.
- There is a possibility of further development in the green supply chain of the model. In this research, the rate of carbon dioxide (CO₂) production from the ambulance is only dependent on the total distance traveled, which can be considered to develop it and bring the model closer to reality, other factors (e.g., speed and traffic).
- Due to each patient's specific type of disease in this study, it can be considered the problem model of nurses with different skills to serve a particular type of disease to bring the problem model closer to reality.
- Other assumptions can be extended to the problem: different services required for each patient and time planning limitations for nurses working to make the research model closer to reality hours.

CRedit authorship contribution statement

Farzin Ziya-Gorabi: Conceptualization, Writing – original draft, Software. **Ali Ghodrtnama:** Supervision, Investigation. **Reza Tavakoli-Moghaddam:** Project administration, Writing – review & editing, Resources. **Mohammad Saviz Asadi-Lari:** Resources, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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